

Bachelor in Computer Applications

Course Title: **C Programming**

Code No: Semester: **1**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Note: Group A (Multiple Choice Questions) is not available from the source. The paper begins with Group B.

Group B

1. What is an expression and operator? Discuss operator associativity in C. Write a program to find factorial of a given integer using recursion.
2. What is an Identifier and Keyword? Explain the rules for naming valid Identifiers in C with example. Write a program to convert a given string to lowercase without using any library functions.
3. What different data types available in C along with their respective range? Write a program to check whether a given integer is palindrome or not.
4. What is the basic structure of a C program? Explain each part.
5. How break statement is different from continue statement? Explain with suitable example.
6. What do you mean by multi-dimensional array? Write a program to display the sum of two m x n matrices.
7. Write a program to create a file "duplicate" that contains the exact copy of file "original".
8. Explain any four input/output functions used in C language with suitable example.
9. Write a program to generate the following pattern of integers.
1121123211234321
10. Write a program to find smallest element of an array using a pointer.
11. Write a program to store and display basic information (Roll, name, address, email, and phone) of students of using a structure.
12. Write Short Notes on:
a. Escape Sequence
b. Null Pointer